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Conference Paper · July 2022

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Abstract**—**Mushroom is one of the important ingredient in our food that has good nutrients. Most types of mushroom are poisonous (inedible), and because of its importance, we need to identify poisonous from eatable mushrooms. Machine learning (ML) techniques such as naïve Bayes, decision tree, SVM, and more applied on mushroom features to classify it into edible or not. There is a limited research on mushroom classification, existed research focus on applying ML techniques individually, where some algorithm perform better in term of accuracy. This research proposed an integrated model that combine most accurate technique's decisions into one decision instead off treating them individually. Mushroom dataset downloaded from UCI repository. Results shows that the performance of the integrated model outperform other techniques by 94% accuracy.

Keywords—machine learning, integradted model, classification algorithm, edible mushroom, inedible mushroom, naïve bayes, decision tree, Support vector machine, Artificial neural networks.

INTRODUCTION

We digest different types of food and focus on healthy food more. Mushrooms are one of the healthiest food on the planet that grow without any efforts. It can grow on/in the ground or on other planets. Mushroom used as an ingredient in most food industry, it has great benefits to our body, where it contains most potent nutrients on the plant such as calcium, phosphorus, vitamins and proteins. Mushroom is used to treat cancer, eradicating viruses, increase immunity system, lose weight, and for good diet programs[1]. Recently, the use of mushroom has been increased by people[2]. However, mushroom can be classified as edible or inedible (poisonous). There are many types of mushrooms and 50-100 types cannot be eaten [1] or we can say that most of mushrooms cannot be eaten[3] and eating collected mushroom directly without knowing it's type is a big mistake and the effect of eating inedible mushroom range from simple symptoms to death.

People is looking at the physical characteristic of the mushroom such as shape, neck length, head diameter, size, color and its environment to decide whether its edible or inedible. Huge amount of mushroom's data collected over years and applications developed to classify mushrooms. Different classification techniques developed and improved to give more accurate decision[4]. The classification algorithms compared for the highest accuracy.

LITRETURE REVIEW

The advanced technology made a huge impact on human life. Different types of tools developed to make life better. Machine learning (ML) is the field of study that enable computers to learn from data[5]. ML used to extract information from data and make decisions. It has been used by all type of industries. There are many machine-learning approaches applied to find the optimal solution from huge data sets. Data classification includes two-step process. The first process is learning step representing by constructing the classification model. The second process is a classification (testing) step and in this step represents the constructed model, which used a given data to predict class labels for them.

The used approaches depends on the nature of the datasets, number of variables, and used model. Supervised learning is an ML algorithm that maps an input to an output based on example input-output pairs. Data will be divided into two sets: training and data sets. The training set is used to create patterns then apply these patterns used to classify the test dataset. The unsupervised approach learns some features from training or previous data then use them to classify new data. K-means is a unsupervised learning algorithms that solve the clustering problem. It defines k centers for each cluster. It is better to have them placed far away from each other.

Decision tree is one of the common method that represents choices and their results in a tree. It has several implementation: ID3, J48, C4.5, Random Forest, Random Tree, ID3+, Oci and Clouds. It has been used to classify the mushrooms whether edible or poisonous based on its behavioral features[4]. They used J48 implementation and the running time on both training and test set is the same with 100% accuracy. A comparative study between most used classification methods[6] shows that the best result obtained by KNN. The KNN used with naïve Bayes to get more accurate and efficient result[7]. naïve Bayes is a classification techniques that based on base theory. Naïve base count the frequencies of data and their values to calculate probabilities. It assumes that a feature in a class is independent of other features[8]. Researches have state that the performance of Bayesian classifier is better than to the performance of decision tree and selected neural network[9]. Image processing techniques with naïve Bayes and KNN algorithms used to classify mushroom[10], naive Bayes shows better accuracy than KNN.

Artificial neural networks (ANNs) are parallel computational models comprised of densely interconnected, adaptive processing units, characterized by an inherent propensity for learning from experience and also discovering new knowledge neural network a to classify whether a mushroom is edible or poisonous[11]. The prediction power of a neural network to classify whether a mushroom is edible or not[12]. The average predictability rate was 99.25%. However, they used the JustNN environment for building the network that was a feed forward Multi-Layer Perceptron with one input layer, three hidden layer and one output layer. A good comparative study shows that SVM out perform all other algorithms by 76% accuracy[13]. The study based on mushroom's texture to determine if it is edible or not. Two phase approach used: training and determination process[9]. Naive Bays and Decision Tree classifiers are used and accuracy of Decision tree is better than Naive Bays, while Naive Bays needs slightly less time than Decision Tree[4].

A mushroom classification model using ML with physical dada of 22 attributes of 800 samples [14]. The study compared several algorithms: Naive Bayes, Naive Bayes, KNN, and SGD Text, where KNN shows the highest accuracy. However, the have run the comparison on 200 samples of edible and inedible. Another study shows that ANFIS performed with bets accuracy[15].

Many other studies has been conducted and the focus was on the accuracy and performance of the individual models. A hybrid model that combine all techniques can be done to achieve better[16].

METHODOLOGY

The main aim of this research is to apply machine-learning approaches to classify mushroom into edible or inedible. Its important to get better decision to avoid side effect of eating inedible mushrooms. We have used a hybrid approach to achieve better accuracy. The proposed models consist of 3 phase: Data pre-processing, classification, and combination

Dataset

Mushroom dataset has been downloaded from UCI mushroom data[17]. The data set composes of 8124 number of rows of data records and 22 attributes. Each mushroom species is identified as class of edible and poisonous. These rows are distributed as 4208 edible mushrooms and 3916 poisonous mushrooms. Table 1 summarizes the attribute, which are used for classifying mushrooms.

Model

The first step is to prepare the data before we proceed more. Prepare the data will include removing null value and repeated features. Python packages used to create dataset from the raw data. The 22 features is important to classify mushroom into edible or inedible. We need to examine these features and decide which one has good contribution in the classification process. We decide to drop two attributes: "stalk-root" and "stalk-root". The reason is that "stalk-root" has 2480 missing values and 'veil-type" feature has only one value and it will not help us in the classification. In addition, it is expected that "gill-color" has the most contribution in the classification and has to be considered.

TABLE 1. Mushroom dataset attribute information

No	Attribute	values
1	cap-shape	bell=b,conical=c,convex=x,flat=f, knobbed=k,sunken=s
2	cap-surface	fibrous=f,grooves=g,scaly=y,smooth=s
3	cap-color	brown=n,buff=b,cinnamon=c,gray=g,green=r, pink=p,purple=u,red=e,white=w,yellow=y
4	bruises	bruises=t,no=f
5	odor	almond=a,anise=l,creosote=c,fishy=y,foul=f, musty=m,none=n,pungent=p,spicy=s
6	gill-attachment	attached=a,descending=d,free=f,notched=n
7	gill-spacing	close=c,crowded=w,distant=d
8	gill-size	broad=b,narrow=n
9	gill-color	black=k,brown=n,buff=b,chocolate=h,gray=g, green=r,orange=o,pink=p,purple=u,red=e, white=w,yellow=y
10	stalk-shape	enlarging=e,tapering=t
11	stalk-root	Bulbous=b, club=c, cup=u, equal=e, rhizomorphs=z, rooted=r, missing=?
12	stalk-surface-above-ring	fibrous=f,scaly=y,silky=k,smooth=s
13	stalk-surface-below-ring	fibrous=f,scaly=y,silky=k,smooth=s
14	stalk-color-above-ring	brown=n,buff=b,cinnamon=c,gray=g,orange=o, pink=p,red=e,white=w,yellow=y
15	stalk-color-below-ring	brown=n,buff=b,cinnamon=c,gray=g,orange=o, pink=p,red=e,white=w,yellow=y
16	veil-type	partial=p,universal=u
17	veil-color	brown=n,orange=o,white=w,yellow=y
18	ring-number	none=n,one=o,two=t
19	ring-type	cobwebby=c,evanescent=e,flaring=f,large=l, none=n, pendant=p, sheathing=s, zone=z
20	spore-print-color	black=k,brown=n,buff=b,chocolate=h,green=r, orange=o,purple=u,white=w,yellow=y
21	population	abundant=a,clustered=c,numerous=n, scattered=s,several=v,solitary=y
22	habitat'	grasses=g,leaves=l,meadows=m,paths=p, urban=u,waste=w,woods=d

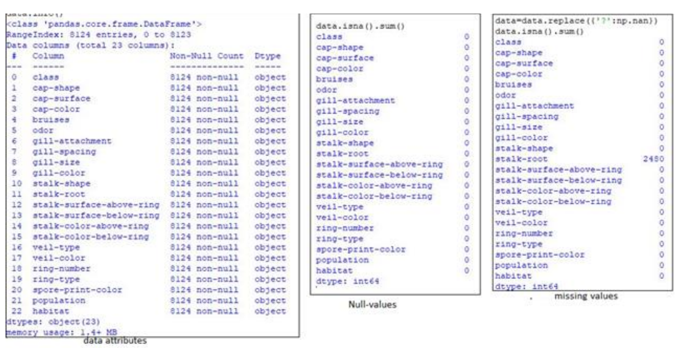

FIGURE 1. Data set view and number of null values

FIGURE 2. Classifiers accuracy comparison

The Second phase is the classification. Several classifiers applied on the data, then the hybrid approach make a decision based on the top three accurate classifiers. We can use more than three but we notice that, it will not change that much in the result. Therefore, the decision on wither mushroom is edible of inedible is taken by combining the decision from most used classifiers. We applied KNN, SVM, naïve base, ANN, logistic regression, and decision tree. The accuracy rounded to closest integer value. KNN outperform all classifiers by accuracy of 94%, where ANN was nt that far by accuracy of 93%. Then SVM with 91% of accuracy. The proposed is perform a little better by combining the decision of KNN, ANN and SVM.

The result of the proposed model is better than individual model performance and it is consistent after running several tests.

CONCLUSION

Mushroom is an important source of essential proteins and vitamins. However, most type of known mushrooms are poisonous. ML learning employed to classify mushrooms into edible or inedible based on its characteristics. Previous research used ML techniques individually, where some techniques perform better than other techniques in term of accuracy. Therefore, it is confusing to decide which methods should we used. In this study, We have applied several machine-learning classifiers on a given dataset "mushroom data". The dataset downloaded from UCI repository. After exploring the data we notice that one feature "stalk-root" contains many missing values and another feature" veil-type" has same values for all rows. These to features are dropped to avoid their effects on the classifications. Where "odor_n" feature is the most important feature that has most effect on the decision. Based on the result we can conclude that if mushroom has odor, it is more like to be inedible.

We notice that most of the classifiers are perform well on the data set, because it is a clean set. However, decision tree, ANN and SVM classifiers outperform the other classifiers. Then we proposed a hybrid model that combine the most top classifiers to improve the accuracy into a hybrid approach. The proposed approach shows better accuracy and consistent results.

The dataset we used is good dataset that does not need to much cleaning and comparing classifiers based on clean data set may not be good way to compare them. Other features may need to be included in order to build and text any model.

FUTURE WORK

It is important to continue this and apply it on several foods. We think about using image-processing techniques to extract mushroom features and apply machine learning to classify it. Deep learning can applied after some times.

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